

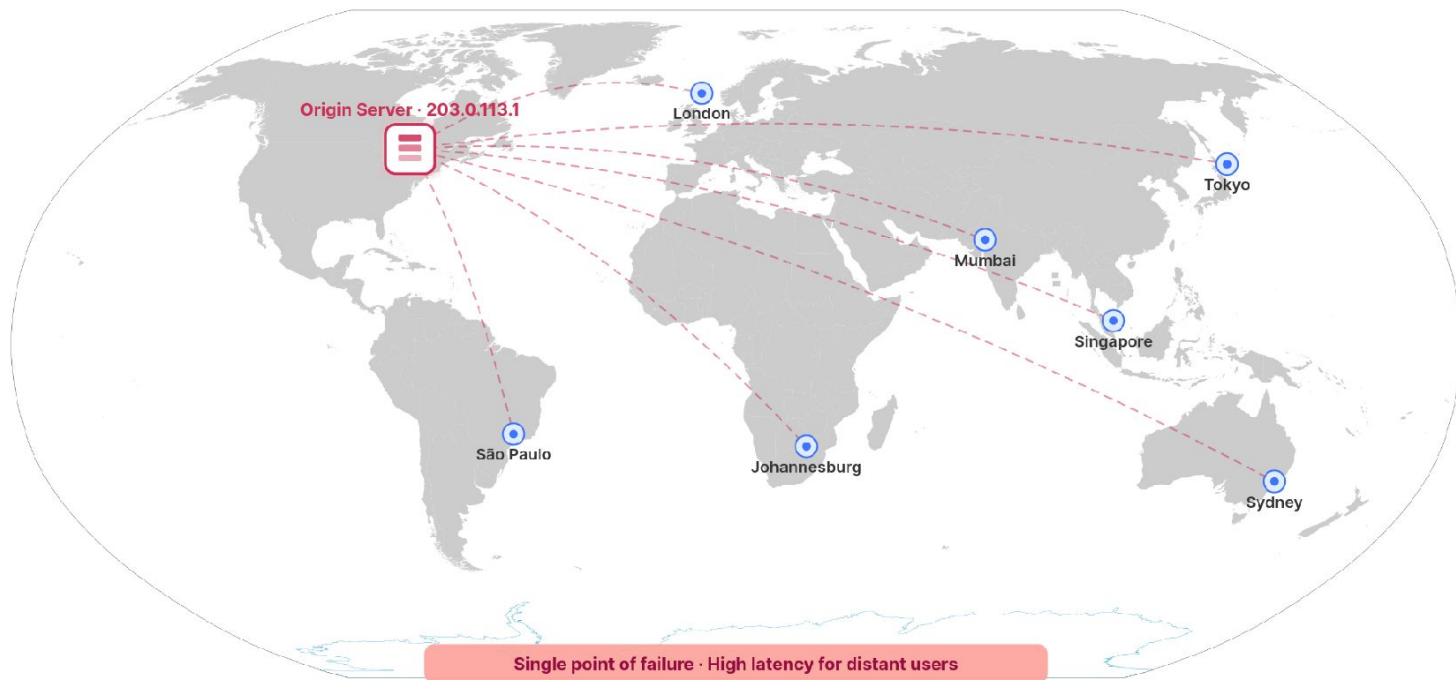
Opportunities and Challenges in AI-driven Anycast Routing Optimization

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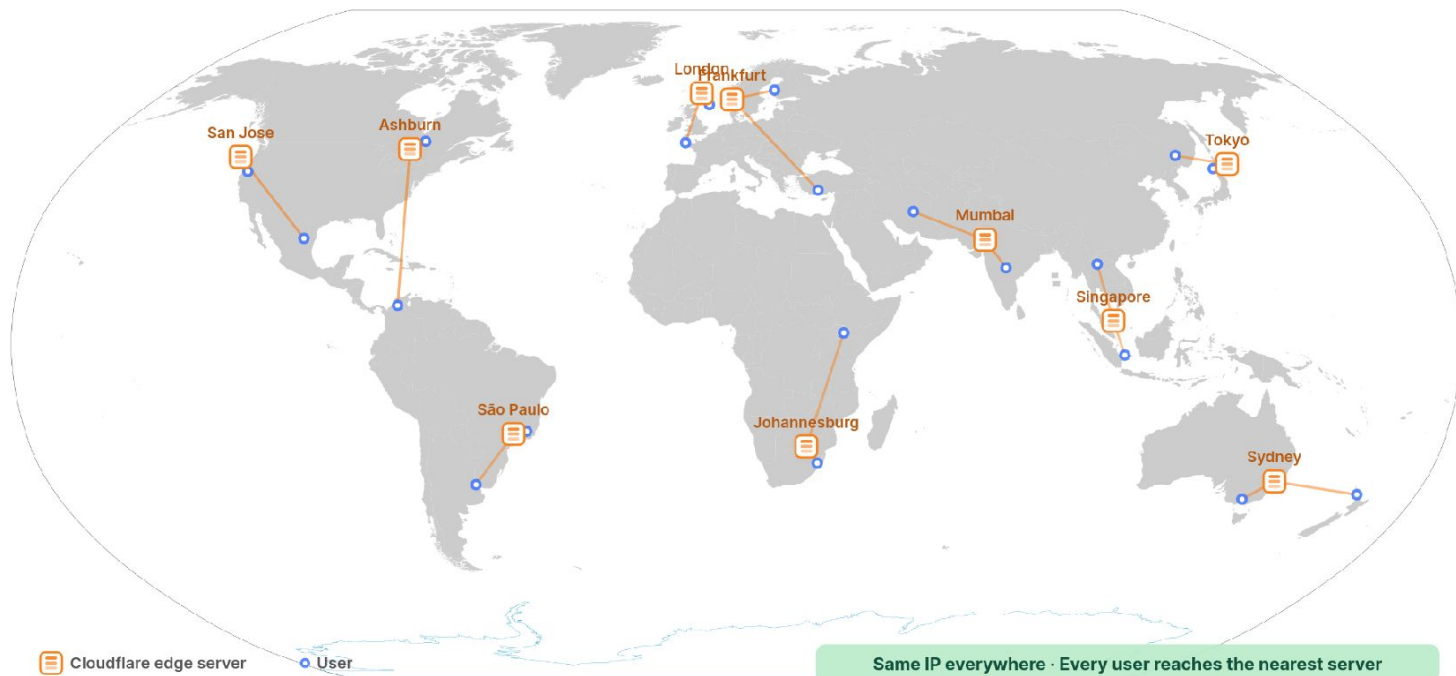
Background: Content Distribution Networks (CDN)

- A **CDN** (content delivery network) is a geographically distributed layer of caching servers that replicates content and answers each request from the replica closest to the client.
- A **PoP (point of presence)** is where edge servers, caches, and routers are colocated and exchange traffic directly with the surrounding networks.
- A CDN's effectiveness depends on its ability to direct clients to its nearest PoP.

Background: Unicast routing



Background: Anycast routing



Anycast Distance Is Measured in Hops, Not Kilometers

BGP: the protocol that routes Internet traffic, has no concept of geography. It simply prefers the route that crosses the fewest networks.

So a user in Nairobi can be routed in Marseille because their ISP's shortest BGP path follows the Internet cables to Europe, even though a geographically closer PoP may exist



Density of PoPs and interconnection helps anycast routing find the nearest PoP to a client

PoPs in 335 cities

Over 13,000 direct
connections between
Cloudflare and ISPs



1. Measuring anycast CDN performance

Measurement goal and methods

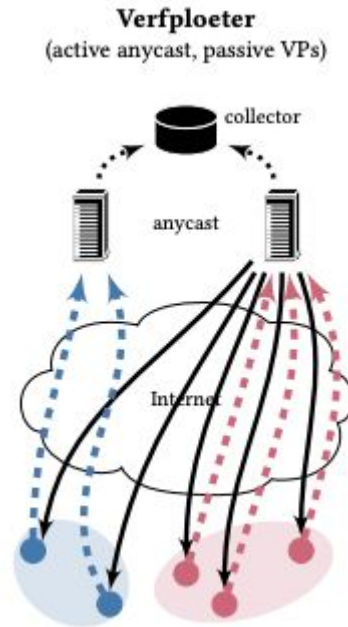
General question: is the client leaded to the PoPs with lowest latency?

General method:

- Mapping each IP to the PoPs that anycast CDN direct it to
- compare the latency between one IP to different PoPs
- So that we can tell whether the latency is "optimized"

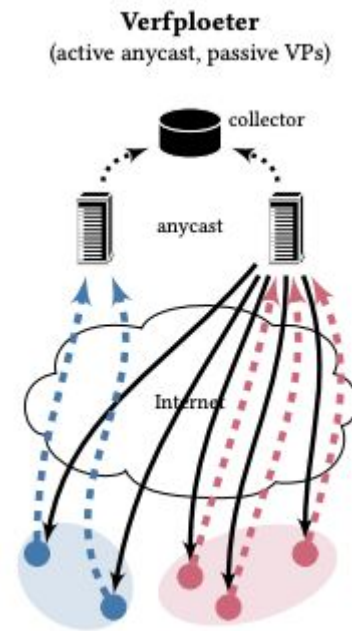
Measurement platform: Verfploeter

- We use the Hypatia measurement platform at Cloudflare, which is based on Verfploeter [0].
 - Verfploeter: a technique to find the catchment of each PoP.
- General idea of Verfploeter:
 - Send ICMP pings from one PoP, with the anycast IP as the source address.
 - Each client's response is routed back by anycast — observe which PoP receives it.
 - The receiving PoP is that client's catchment PoP.



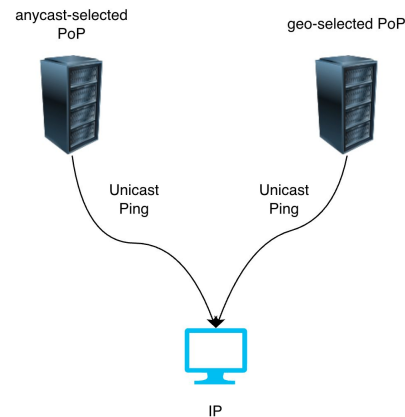
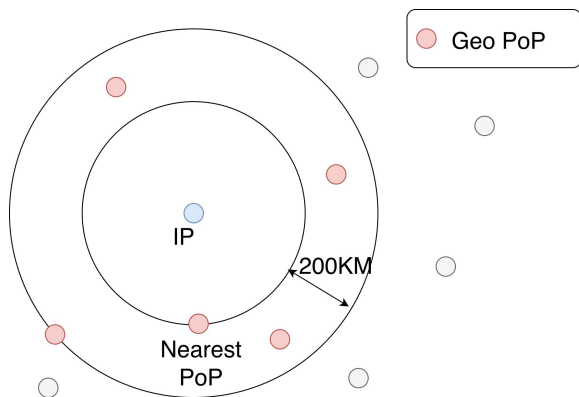
Phase 1: Anycast Ingress Mapping

- For every client IP in the target hit list:
 - Send Anycast ICMP probes from several main PoPs to identify the active ingress PoP.
 - Use unicast probe to measure the latency between client IP and anycast-selected PoP.
- **Goal:** For each client IP, find which PoP is selected by anycast, and RTT.



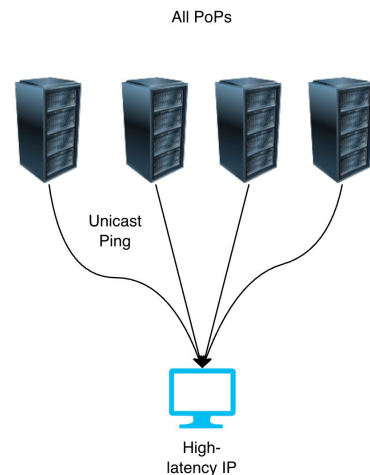
Phase 2: Geo-nearby PoPs measurements

- For each IP, find geo-nearby PoPs
 - Geolocate IPs using IPInfo dataset
- Send unicast probe to measure RTT between client IPs and Geo-nearby PoPs
- **Goal:** For each client IP, find Geo-nearby PoPs and measure RTTs



Phase 3: Full scanning for high-latency IPs

- For anomalous IPs exhibiting high latency across **both** the Anycast-selected PoP and the Geo-selected PoP:
 - Trigger full-mesh Unicast probes from all available PoPs to uncover hidden, low-RTT routing paths.
- **Goal:** For high-latency client IP, check if there are any optimal PoPs which are not selected by anycast or geolocation.



Results

- **Hitlist:** 12.2M target IPs, sampled from a global responsive-IP hitlist (at most 5 IPs per /24 prefix).
- **Measurement interval:** the full scan pipeline is re-executed every 12 hours, to capture longitudinal routing fluctuations.
- **Duration:** 10 snapshots accumulated for longitudinal analysis.

Finding 0: Anycast Matches or Beats Geo-Best for ~91% of IPs

- Geo-best: the lowest-RTT PoP among each IP's geo-nearby PoPs.
- For **58.7%** of IPs anycast selects exactly the geo-best PoP.
- **41.3%** select a different PoP, but among these:
 - 51.6% tie ($\Delta\text{RTT} \leq 10$ ms) · 26.5% anycast wins · 21.9% geo wins
- Overall: anycast is as good as or better than geo-best for ~91% of IPs; only ~9% would benefit from switching.

Case Study: When Anycast Beats Geography



Case Study: CN (IP level) — Anycast Beats the Geo-Nearest PoP

Client IP: 117.141.130.79 (Nanning, CN — China Mobile)
Geo-nearest PoP: han-c (Hanoi), 314 km — 301 ms
Anycast PoP: hkg-a (Hong Kong), 578 km — 28 ms
Anycast wins by 273 ms — consistent across the prefix

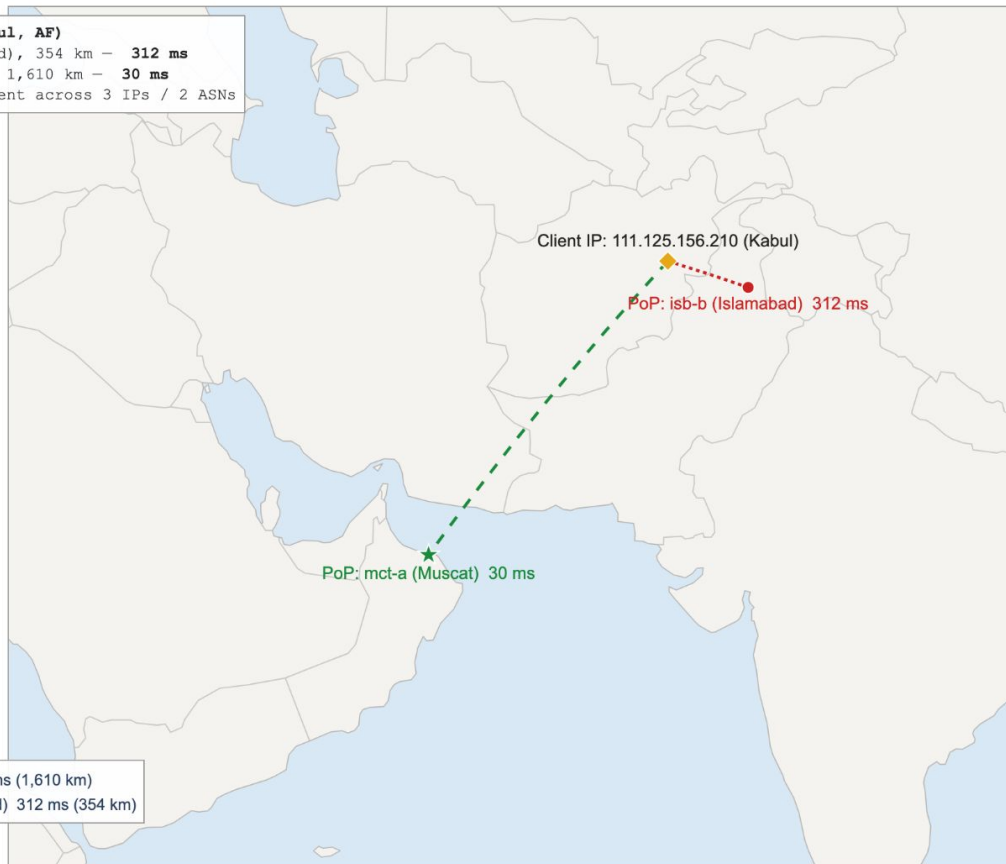


Case Study: When Anycast Beats Geography



Case Study: AF (IP level) — Anycast Beats the Geo-Nearest PoP

Client IP: 111.125.156.210 (Kabul, AF)
Geo-nearest PoP: isb-b (Islamabad), 354 km — 312 ms
Anycast PoP: mct-a (Muscat), 1,610 km — 30 ms
Anycast wins by 282 ms — consistent across 3 IPs / 2 ASNs



Finding 1: Not all Anycast PoPs are Equal

Top N PoPs	IP Volume	Ratio (%)
Top 5	~1.47M	23.2%
Top 10	~2.41M	38.0%
Top 20	~3.63M	57.3%
Top 50	~5.11M	80.7%
Top 100	~5.91M	93.4%

Table. Share of analysis IPs attracted by the top-N PoPs.

- The top 20 PoPs (out of 501 PoPs) serve 57% IPs

Finding 2: Anycast PoP Assignment is Stable

- Across 10 snapshots, 5,918,222 IPs (94.5%) consistently selected the same PoP.
- The remaining 5.5% experienced PoP flipping.

Case Study 1: Inter-Region Path Inflation IP: 103.151.23.73

Snapshot	PoP	RTT (ms)
scan_20260605_210246	sin-a	13.1
scan_20260606_042714	sin-a	14.6
scan_20260606_122714	cgk-a	999.4 (Catchment Shift)
scan_20260606_202714	sin-a	13.2 (Path Reversion)
scan_20260607_042714	sin-a	14.8

Unique PoPs: 2 | RTT Inflation Delta: +986.3 ms

Case Study 2: Intra-Continental Sub-optimal Shift IP: 205.178.136.101

Snapshot	PoP	RTT (ms)
scan_20260605_210246	atl-a	1.1
scan_20260606_042714	mia-a	13.7 (Catchment Shift)
scan_20260606_122714	atl-a	1.2 (Path Reversion)
scan_20260606_202714	atl-a	1.5
scan_20260607_042714	atl-a	1.4

Unique PoPs: 2 | RTT Inflation Delta: +12.6 ms

Case Studies: PoP Flipping and Its RTT Impact

Finding 3: Flipping Causes Small RTT Shifts, but with a Long Tail

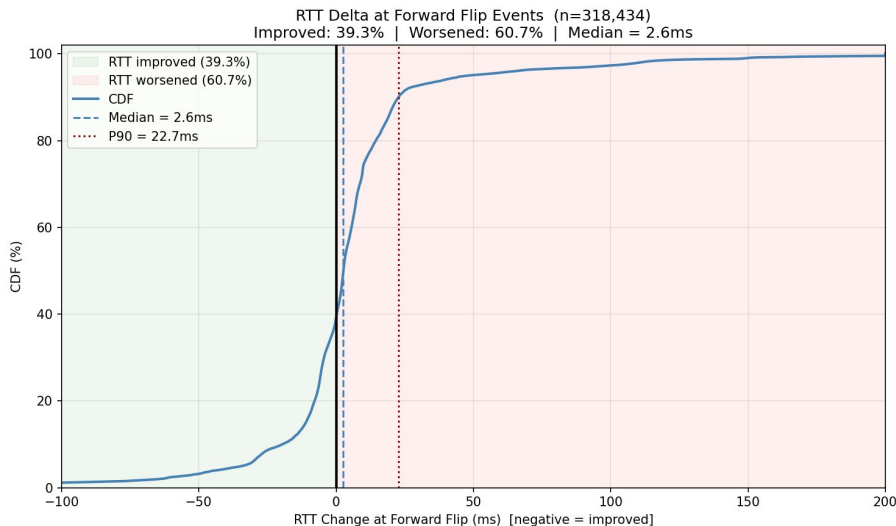


Fig. CDF of RTT change per flip event. Positive values mean RTT got worse after the flip.

- 60.7% of flip events worsen RTT.
- Most shifts are small: 90% are within 22.7 ms.
- But the tail is long: extreme flips inflate RTT by more than 100 ms, severely degrading user experience.

Finding 4: the Nearest is Not Always the Fastest

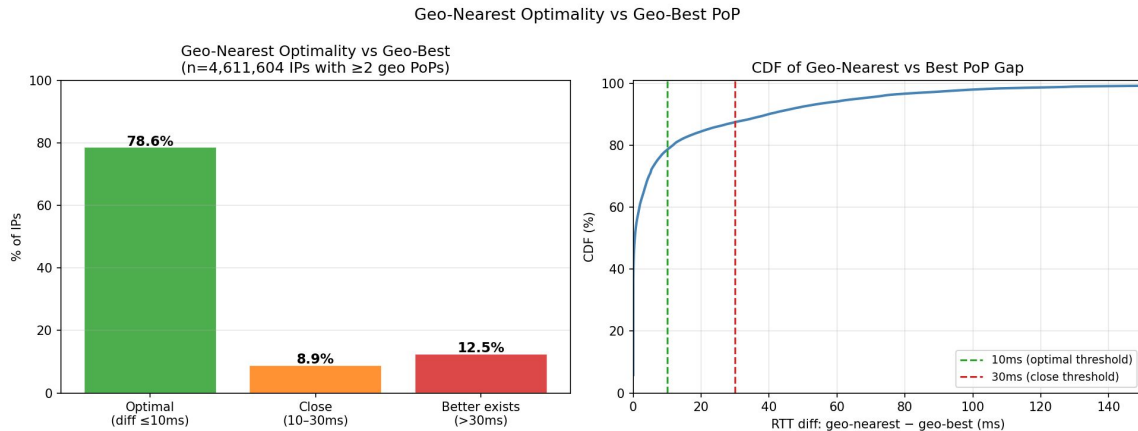


Fig. RTT advantage of the fastest geo-nearby PoP over the geo-nearest PoP (IPs with ≥ 2 geo-nearby PoPs).

- **Recall:** we probe all PoPs within +200 km of each IP's geo-nearest PoP.
- Among 4.6M IPs with ≥ 2 geo-nearby PoPs:
 - 12.5% have a non-nearest PoP that is faster by >30 ms, another 8.9% faster by 10–30 ms
- **Takeaway:** physical proximity does not reliably predict latency — for ~21% of IPs, the fastest nearby PoP is not the nearest one.

Finding 6: Neither Anycast-Selected nor Geo-Nearby is the Best for High-Latency IPs

Table. Best PoP for high-latency IPs. RTT saving is relative to the anycast-selected PoP.

Best PoP for the IP	Share of high-latency IPs	Median RTT saving
Anycast-selected PoP	14.5%	— (baseline)
Geo-nearby PoP	6.0%	25.88 ms
Other PoP	79.5%	36.91 ms

- **Scope:** high-latency IPs — RTT > 50 ms to both the anycast-selected and geo-nearby PoPs (6.6% of all IPs), probed from all PoPs.
- For 79.5% of high-latency IPs, the best PoP is one that neither BGP nor geography would pick — switching saves a median of 36.9 ms.
- **Takeaway:** for the high-latency tail, the real gains lie beyond the nearby PoPs.

Case Study: Neither BGP nor Geography Found the Fast PoP



Finding 7: Suboptimal client IPs concentrate to few PoPs

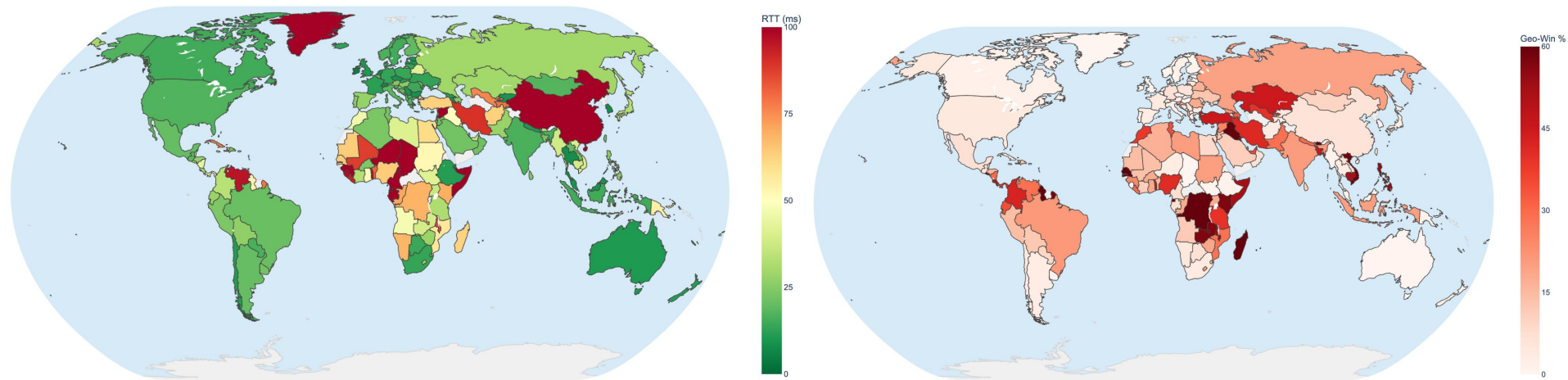


PoP	Suboptimal IPs	% of all suboptimal IPs	Median / P90 saving
sin-a (Singapore)	184,004	20.4%	27 / 41 ms
lax-a (Los Angeles)	96,928	10.8%	125 / 141 ms
hkg-a (Hong Kong)	66,908	7.4%	26 / 50 ms
arn-a (Stockholm)	41,853	4.6%	18 / 21 ms
fra-a (Frankfurt)	31,478	3.5%	28 / 73 ms
Top 5 combined	421,171	46.8%	—

Table. Top 5 PoPs by absorbed suboptimal IPs (threshold: >10 ms saving available).

- For each PoP, we count how many IPs are suboptimal to it (an IP is suboptimal if some other PoP is >10 ms faster than its anycast PoP).
 - 14.2% of all IPs (900K) are suboptimal.
 - 381 of 435 PoPs absorb at least some suboptimal IPs.
 - The top 5 absorbers alone take ~47% of all suboptimal IPs.

Finding 8: Higher Improvement Potential Cluster in Asia and Africa



Left Fig: median anycast RTT by region. Right Fig: median RTT saving if switched to geo-best PoP.

- Higher anycast latency: concentrated in China, Central Asia, parts of Africa and the Middle East.
- Geo-routing improvement potential: highest in Central/East Africa, Central Asia, parts of Latin America.
- The two maps do not overlap perfectly — high latency $\neq$ high fixability.

2. Room for optimization, and is AI the solution?

Oracle: If all IPs can selected best PoP

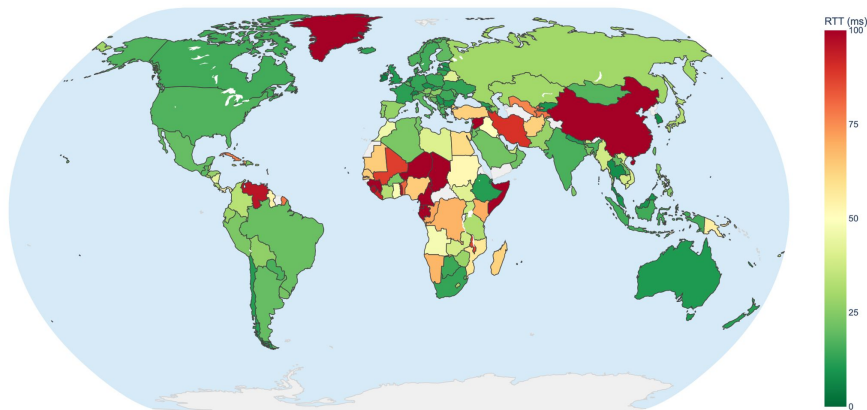


Fig. RTT distribution under current anycast.

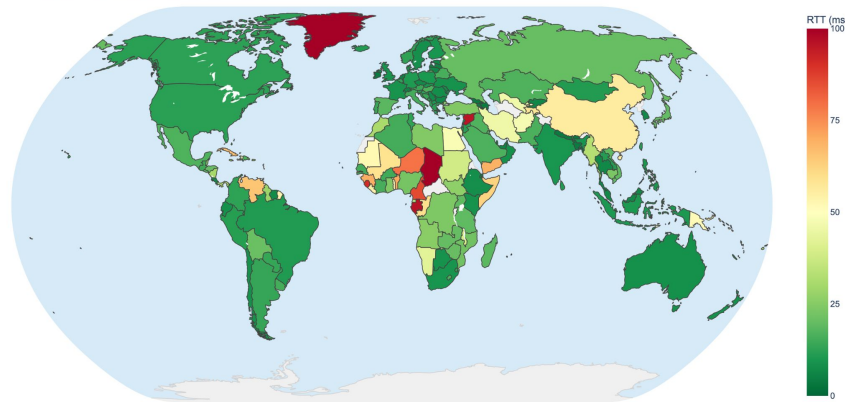
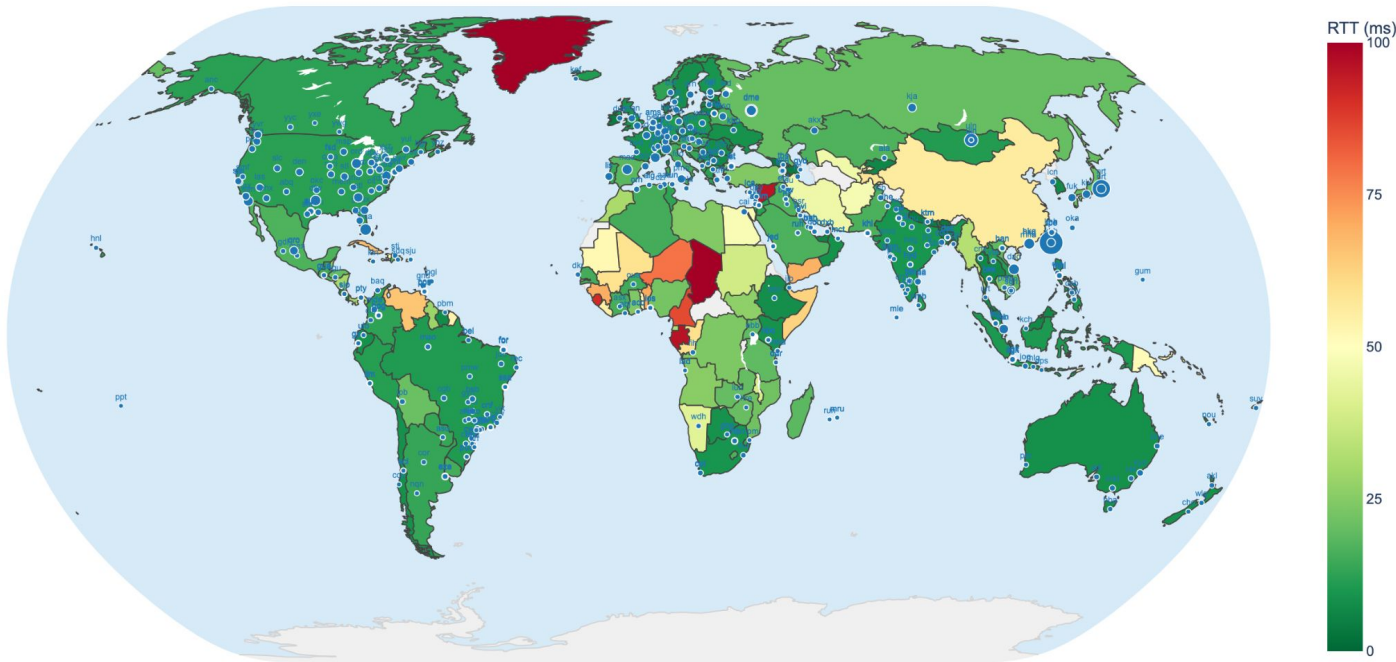


Fig. RTT distribution if every IP were served by its best PoP (oracle).

- **Oracle:** If Every IP Could Select Its Best PoP
 - RTT would improve for 14.2% of IPs ($> 10\text{ms}$).
 - P90 RTT across all IPs would drop by 20.3 ms.

Further Optimization: Why Building More PoPs Might not Fix the Edge

Fig. Per-country RTT under global-best routing; blue dots = active PoPs (size $\propto$ IPs served). Countries that stay red have PoPs right next to them — the path, not the distance, is the bottleneck.



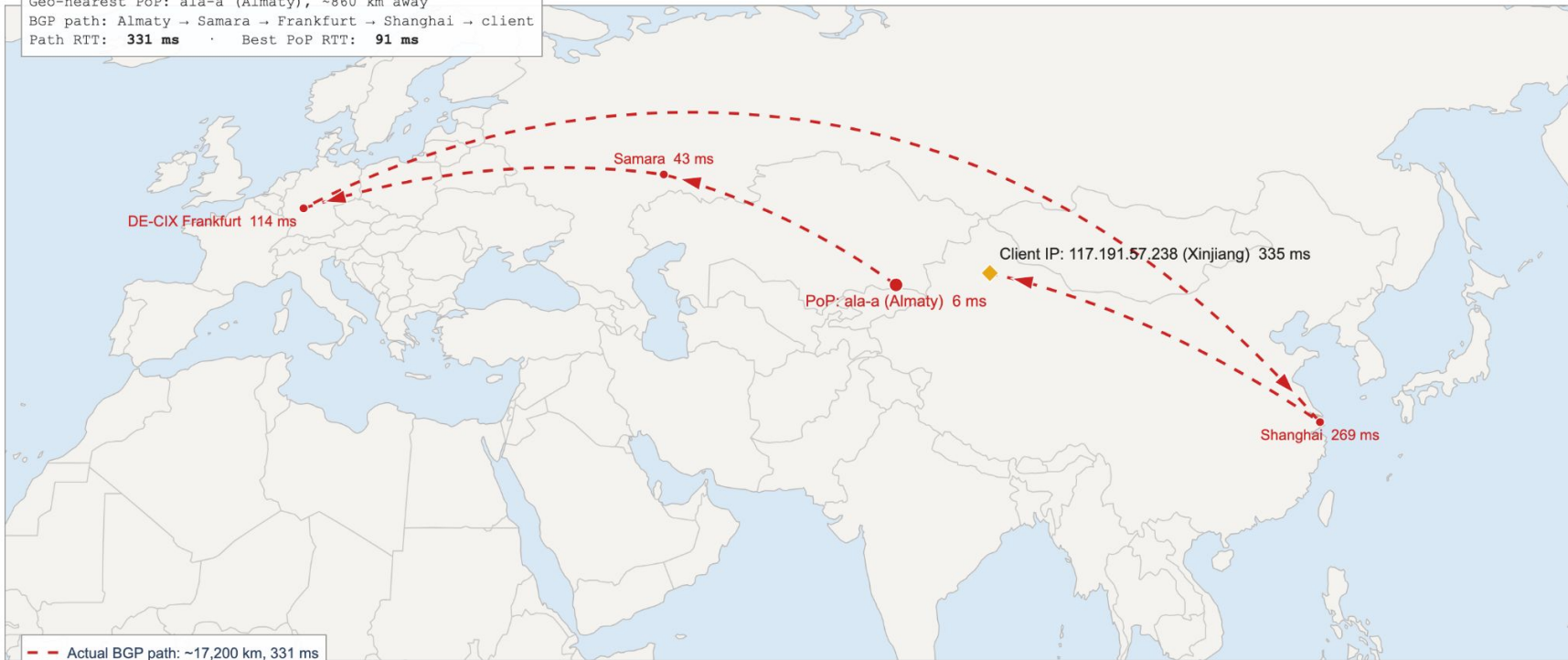
- Under the global-best oracle, high latency persists in West/Central Africa, the Caribbean, and Central Asia.
- These regions are not unserved — healthy PoPs sit inside or right next to the slow zones.
- Fixing this needs control-plane tuning (audit BGP links, adjust peering), not blind PoP expansion — a new PoP acts like a magnet, pulling in traffic it was never meant to serve.

Case Study: When Anycast Beats Geography



Case Study: Geographic Detour — a ~860 km Neighbor, 331 ms Away

Client IP: 117.191.57.238 (Xinjiang, CN)
Geo-nearest PoP: ala-a (Almaty), ~860 km away
BGP path: Almaty → Samara → Frankfurt → Shanghai → client
Path RTT: 331 ms · Best PoP RTT: 91 ms



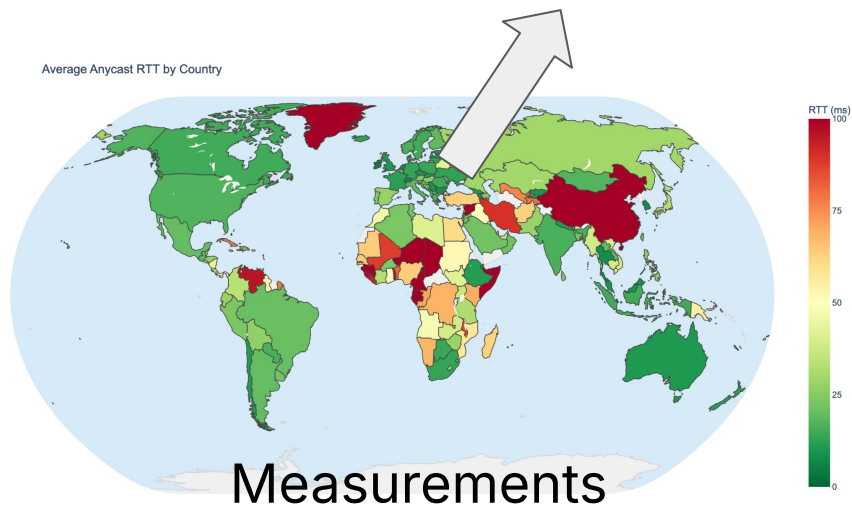
Optimize BGP anycast

Some examples:

- **AS-path prepending:** Make selected routes appear less attractive by artificially increasing the AS-path length.
- **BGP communities:** Ask upstream providers to change route propagation, local preference, or geographic scope.
- **Local preference:** Prefer specific routes within an operator or provider network, typically based on policy rather than performance.
- **Selective announcements:** Announce a prefix only from selected sites or to selected providers.

Challenges for anycast optimization

1. Where needs optimization?

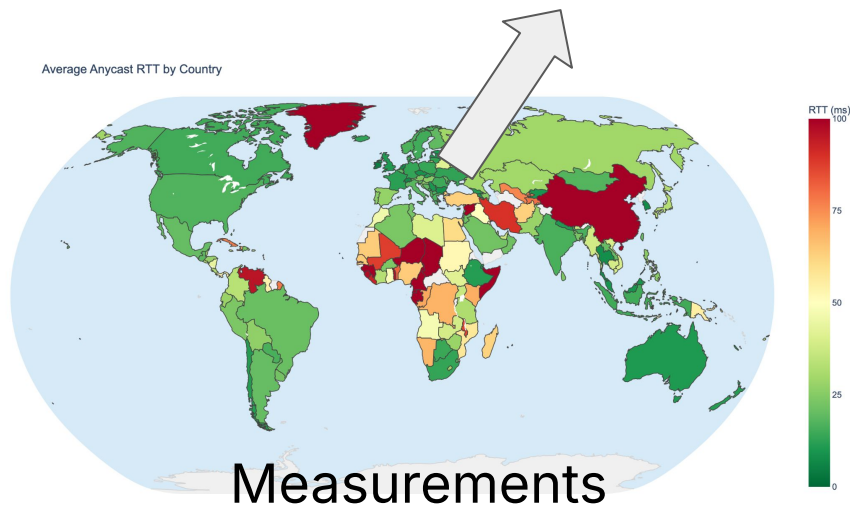


Challenges for anycast optimization

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2. Where to optimized?

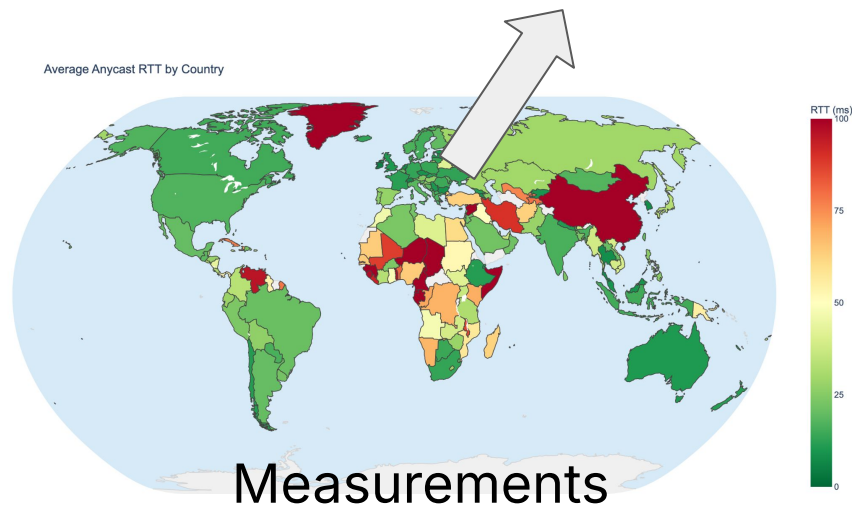


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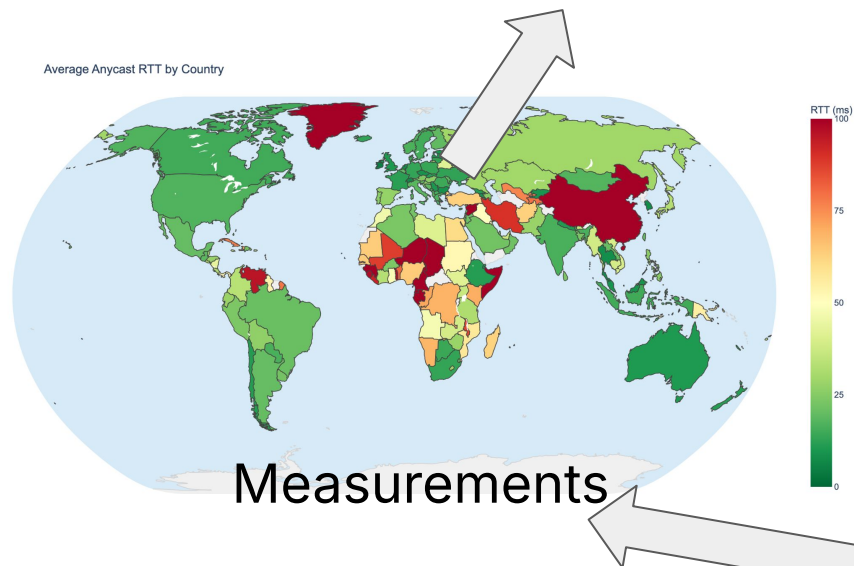
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Challenges for anycast optimization

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2. Where to optimize?



3. how to optimize?

Previous papers in Anycast Optimizing

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- **Internet Anycast: Performance, Problems, & Potential** — Zhihao Li et al. (SIGCOMM 2018)
- **Idea:** Attach a **self-designed BGP community tag** (encoding each site's geo-location) to anycast ads, so routers steer traffic to the geographically nearest PoP.

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- **Idea:** Attach a **self-designed BGP community tag** (encoding each site's geo-location) to anycast ads, so routers steer traffic to the geographically nearest PoP.
- **Con:** Needs voluntary ISP adoption; only target at geo-nearest PoP.

Existing Work in Anycast Optimizing

- **AnyOpt: Predicting and Optimizing IP Anycast Performance** — Xiao Zhang et al. (SIGCOMM 2021)
- **Idea:**
 - Optimize anycast by turning off some PoPs/providers.
 - Open two PoPs at a time, and see which PoP is selected by each client
 - Optimize by choosing which PoPs to turn on/off to minimize total client latency.

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 - Open two PoPs at a time, and see which PoP is selected by each client
 - Optimize by choosing which PoPs to turn on/off to minimize total client latency.
- **Con:** pairwise BGP tests grow quadratically with number of PoPs
 - a layered optimization helps, but gives little benefit when many sites sit behind few providers.

Existing Work in Anycast Optimizing

- **AnyPro: Preference-Preserving Anycast Optimization based on Strategic AS-Path Prepending** — Minyuan Zhou et al. (NSDI 2026)
- **Idea:**
 - Optimize by tuning the AS-path prepending at each PoP.
 - General idea: Change prepending length at some PoPs, measure the catchment changes, use measurement data to determine the final prepending length of each PoP.

Existing Work in Anycast Optimizing

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- **Idea:**
 - Optimize using AS-path prepending at each PoP.
 - General idea: Artificially change path length at some PoPs, measure the catchment changes, use measurement data to determine the final prepending length of each PoP.
- **Con:** Only works for ASPP-sensitive clients; conflicting client preferences must be resolved and can't all be satisfied; expensive measurement.

Could AI help?

- Continuously run active and passive network measurements.
- Generate candidate BGP configurations automatically.
- Predict the impact of different routing configurations.
- Recommend routing actions under multiple objectives, such as latency, load balancing, reliability, and cost.
- Continuously adapt routing policies as traffic demand and network conditions change.
- Detect unexpected outcomes and automatically trigger rollback.

Could AI help?

Goal: An Automated Anycast Optimization Pipeline

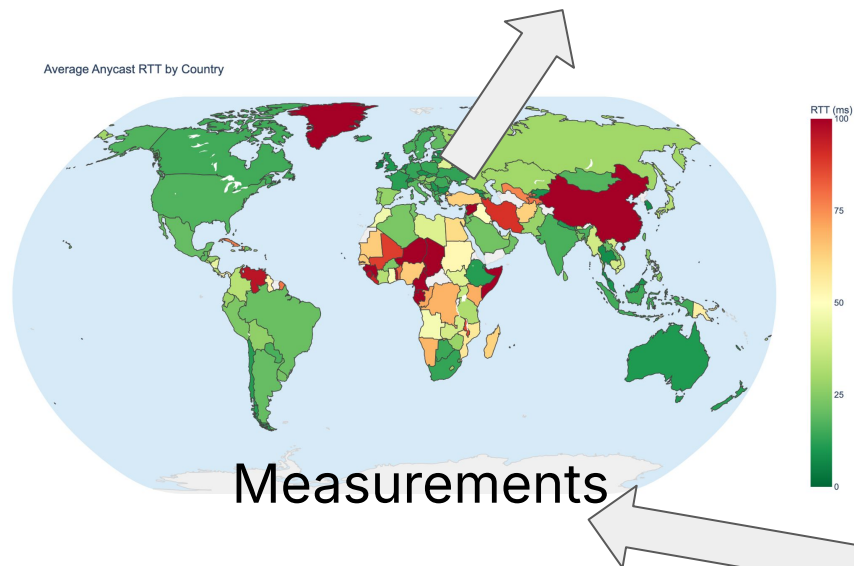
Measurements → Prediction → Optimization → Configuration Generation
→ Safe BGP Action → Validation → Rollback

Could AI help?

1. Where needs optimization?



2. Where to optimize?



3. how to optimize?

**Thank you
Questions?**